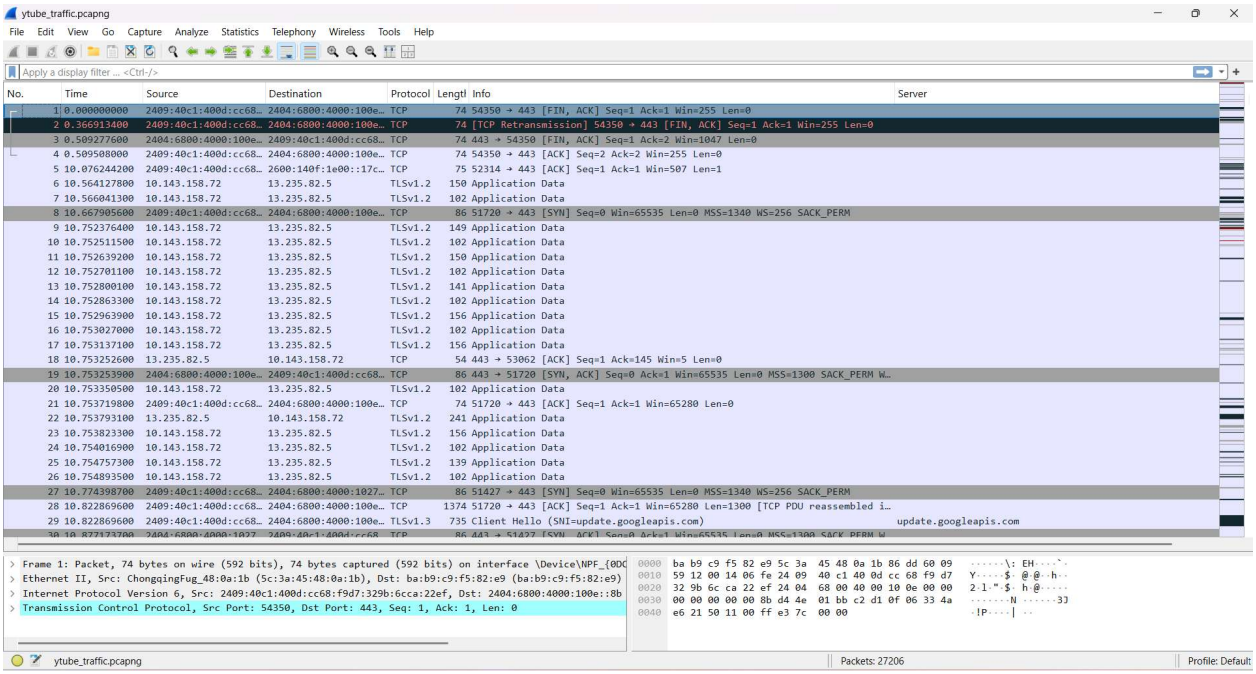
3: Using Advanced Tools Theory

Task 1: Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as insta_traffic.pcap.

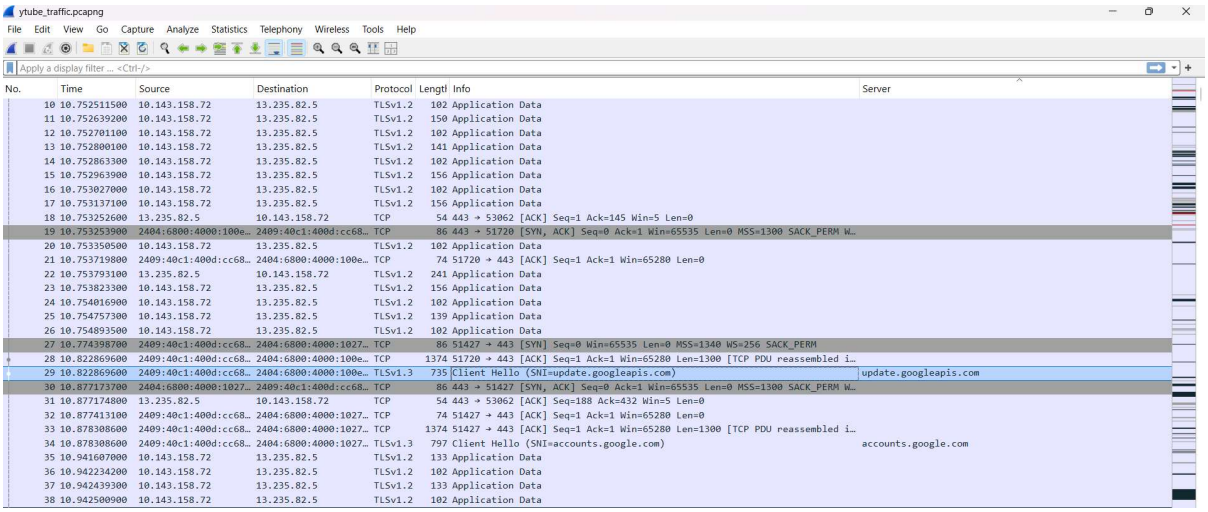
1. Installed Wireshark on Windows 11 together with Npcap, the packet capture driver.
2. Closed all background applications and paused Windows Update to reduce unwanted traffic.
3. Opened Wireshark, selected the Wi-Fi interface and started the capture.
4. Opened a new Incognito window in Chrome and browsed YouTube.
5. Stopped the capture with the red stop button and saved it with File > Save As.

Capture file saved: ytube_traffic.pcapng (27,206 packets captured).

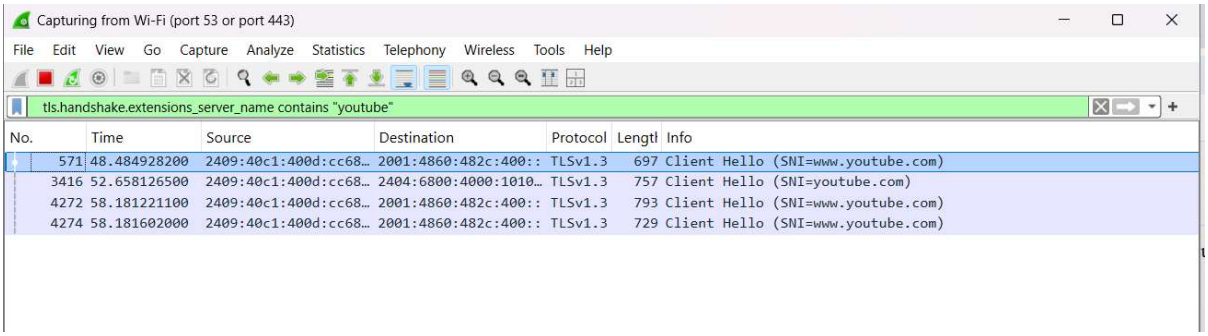


Task 2: Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.

A new column named Server was added (Edit > Preferences > Appearance > Columns, field tls.handshake.extensions_server_name) so the domain name is visible on every Client Hello row.



Server column showing the domain in each Client Hello packet (for example update.googleapis.com and accounts.google.com).



Filter on TLS server names containing "youtube" showing Client Hello packets to www.youtube.com.

The printed capture contained 94 filtered packets, and a server name could be read from 89 of them. Counts grouped by main domain:

Main domain	Connections	What it is
google.com	19	Search, accounts, Play and ads services
googleapis.com	15	Background Google services (Chrome updates, sign-in)
gstatic.com	14	Google static files (fonts, thumbnails)
ytimg.com	6	YouTube thumbnails and images
youtube.com	6	YouTube site (www, accounts, search suggestions)
googlevideo.com	5	YouTube video streaming servers
gvt2.com	4	Google update and beacon traffic
googleusercontent.com	4	User and profile images

Most frequent individual domains (4 connections each): www.google.com, update.googleapis.com, play.google.com, i9.ytimg.com, encrypted-tbn3.gstatic.com, beacons.gcp.gvt2.com.

Observation: Almost all connections go to Google-owned domains. A YouTube page loads from many supporting domains, and the actual video comes from googlevideo.com servers. A few entries

(quillbot.com, stream-production.avcdn.net) are not YouTube and came from other open tabs or applications on the laptop.

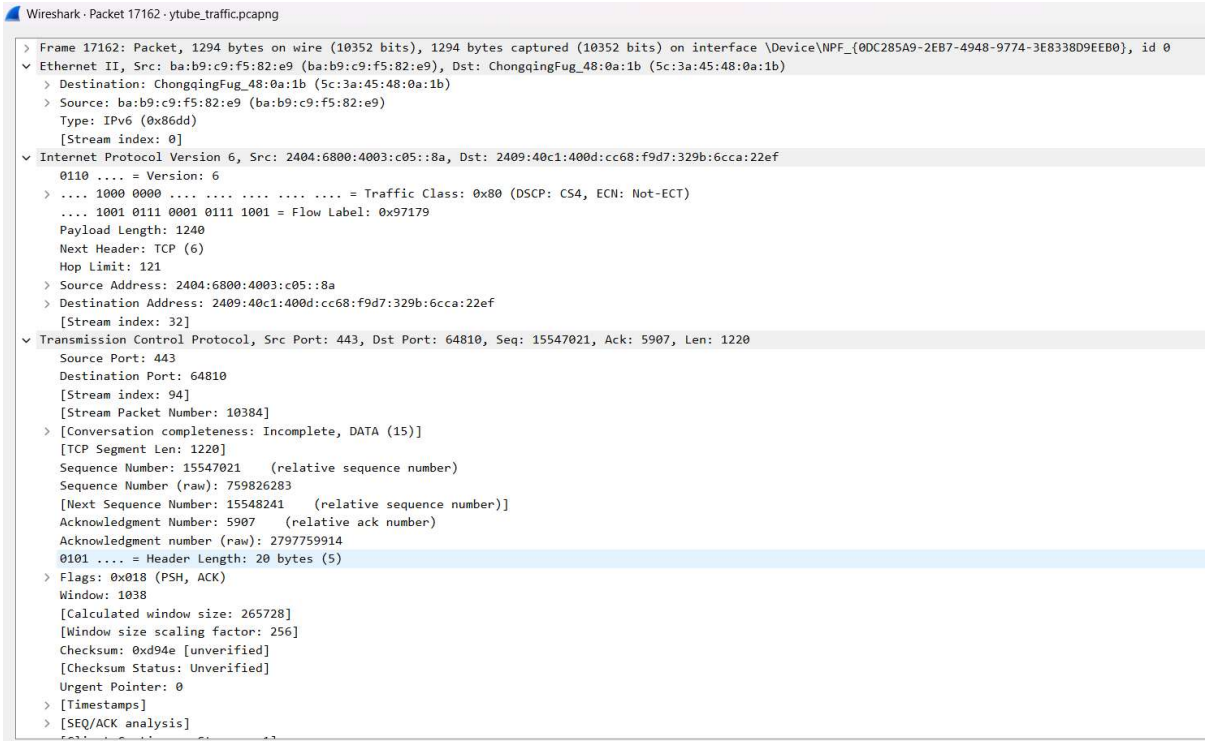
Task 3: Open your insta_traffic.pcap in Wireshark and find a TCP packet. Document the source IP, destination IP, and TCP port numbers for that packet.

Opened the saved capture (ytube_traffic.pcapng), applied the display filter tcp and selected packet 17162. The Internet Protocol Version 6 and Transmission Control Protocol sections were expanded in the packet details pane.

```
tcp
```

Field	Value
Packet number	17162
Source IP	2404:6800:4003:c05::8a
Destination IP	2409:40c1:400d:cc68:f9d7:329b:6cca:22ef
Source port	443
Destination port	64810
IP version	IPv6
TCP flags	PSH, ACK (0x018)
TCP payload length	1220 bytes (frame length 1294 bytes)

The source port 443 is the standard HTTPS port, so this packet was sent by a Google server to the laptop. The laptop used the temporary port 64810 for this one connection.



Packet 17162 details showing the IPv6 addresses and TCP source and destination ports.

Task 4: Run an Nmap scan on your own machine (localhost) to detect open ports. Save the command you used and list the open ports found.

Command used

```
nmap -A -v 127.0.0.1
```

Nmap was installed from nmap.org and run from Command Prompt. The -A option enables service version detection, OS detection and default scripts, and -v gives verbose output.

Open ports found (6 open, 994 closed)

Port	Service	Details
21/tcp	ftp	Microsoft ftpd (FTP file transfer server)
80/tcp	http	Microsoft IIS httpd 10.0, page title "IIS Operation Test Page"
135/tcp	msrpc	Microsoft Windows RPC (core Windows service)
445/tcp	microsoft-ds	SMB, Windows file sharing
5357/tcp	wsdapi	Web Services for Devices (Windows device discovery)
8080/tcp	http	Microsoft IIS httpd 10.0, page title "IIS Cache & Compression Tester"

Key output

```
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00099s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE      VERSION
21/tcp   open  ftp          Microsoft ftpd
80/tcp   open  http         Microsoft IIS httpd 10.0
135/tcp  open  msrpc        Microsoft Windows RPC
445/tcp  open  microsoft-ds?
5357/tcp open  wsdapi?
8080/tcp open  http         Microsoft IIS httpd 10.0
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
Nmap done: 1 IP address (1 host up) scanned in 31.73 seconds
```

Findings

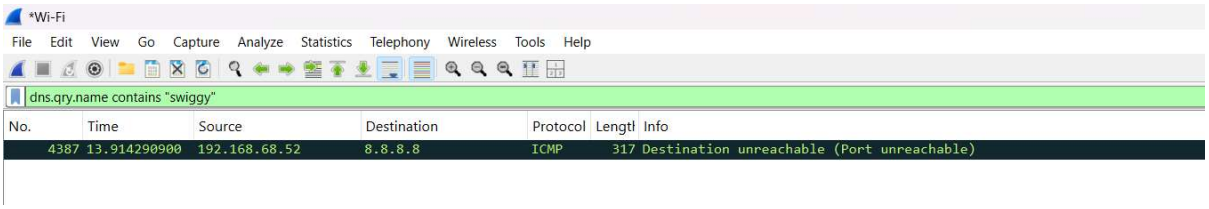
- The operating system was detected as Windows.
- Ports 21, 80 and 8080 show that an FTP server and two IIS web servers are running on the laptop.
- Nmap flagged the HTTP TRACE method as potentially risky on both web servers.
- Ports 135 and 445 are normal on.

Task 5: Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved.

To force a fresh DNS lookup, the Windows DNS cache was cleared with ipconfig /flushdns and the Chrome host cache was cleared.

A new capture was started on the Wi-Fi interface and swiggy.com was opened in an Incognito window in the browser. The display filter used was:

```
dns.qry.name contains "swiggy"
```

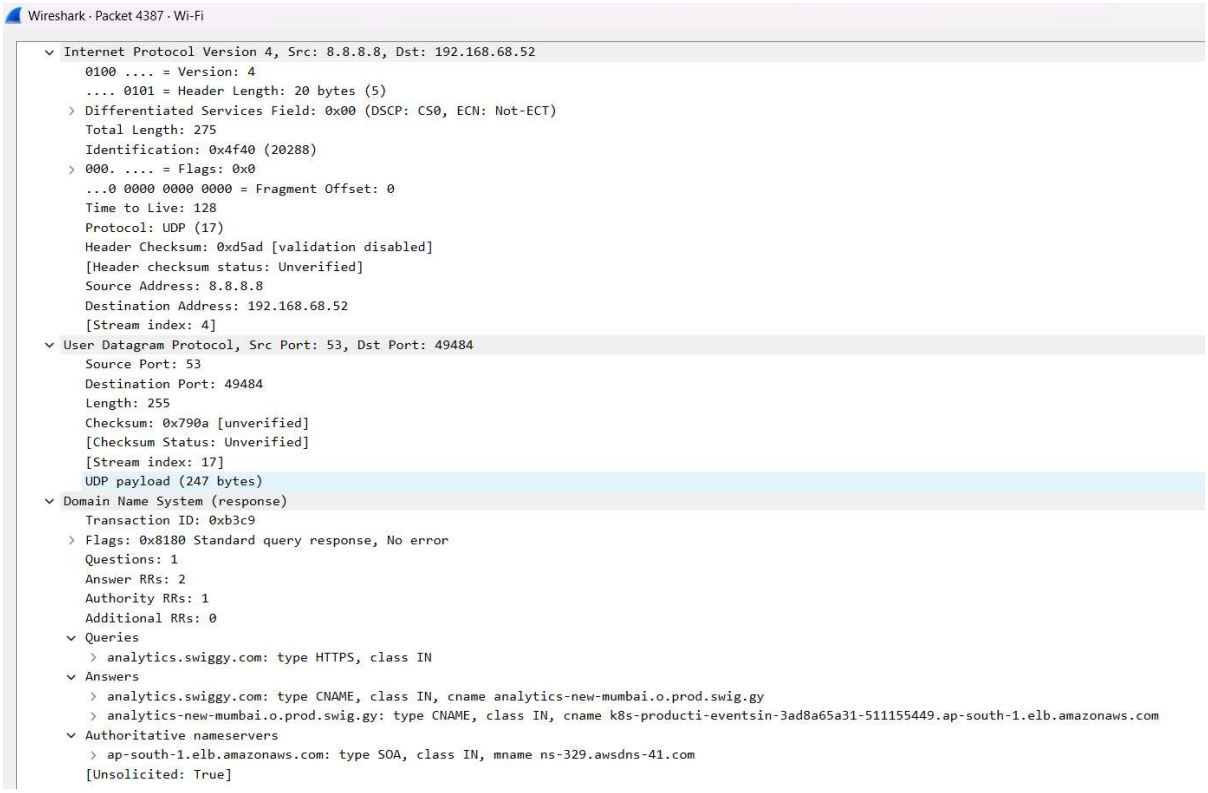


Display filter for Swiggy DNS traffic applied in Wireshark (packet 4387).

Packet 4387 was opened and the Domain Name System section was expanded:

Field	Value
Packet number	4387
Domain name being resolved	analytics.swiggy.com
Record type / class	HTTPS (type 65), class IN
DNS server	8.8.8.8 (Google Public DNS), source port 53
Laptop	192.168.68.52, destination port 49484
Transaction ID	0xb3c9
Result	Standard query response, No error

Thepaket contained: analytics.swiggy.com is an alias (CNAME) for analytics-new-mumbai.o.prod.swig.gy, which is an alias for an Amazon load balancer in the ap-south-1 (Mumbai) region (k8s-producti-eventsin-...ap-south-1.elb.amazonaws.com). Swiggy's analytics service therefore runs on Amazon Web Services in Mumbai.



DNS response for packet 4387 showing the queried domain analytics.swiggy.com and the CNAME answers.